

Regulatory-Ready Platform

for Multiomics Diagnostics

COMMERCIAL BRIEFING

From discovery scripts to auditor-ready evidence — without rebuilding your stack for every assay.

The problem buyers already feel

- Biomarker → authorization (**510(k)** / **De Novo** / **PMA**, plus **CLIA**/LDT) is historically **slow**.
- Validation fails quietly: train/holdout **leakage**, ad hoc notebooks, unreproducible “final” runs.
- Audits bottleneck on **documentation**, not just science.
- Every new omics modality or indication tends to become a **new product stack**.

MethylPipeline is built to shrink those failure modes — with config-first science, partition discipline, and submission scaffolds.

Claim boundary (say this early)

The platform **enforces process and configuration controls** and **assembles evidence packages**. Architecture, SaMD profiles, and filled feasibility packages are **not** FDA clearance or approval.

- Novel diagnostics may need **De Novo** or **PMA**, not only a predicate **510(k)**.
- Laboratory deployment may also implicate **CLIA**.
- We shorten the **software + validation process** path — we do **not** choose or guarantee the regulatory pathway.

Positioning

Regulatory-Ready Platform for Multiomics Diagnostics

Bridges **bioinformatics R&D engineering** and **regulatory / quality readiness**.

Three pillars of value

1. R&D agility

New cohorts, modeling backends (ECDF, tabular sklearn, covariates), and cell-type estimation (Houseman / HiTIMED) via **configuration** — not ad hoc scripts.

2. Cloud efficiency

Cluster-parallel workers + **CAAS** skip redundant alignment/extraction when only downstream science changes — lower VM cost on large cohorts.

3. SaMD controls

Guard against statistical contamination, stage clinical-performance claims, and auto-assemble regulatory **evidence scaffolds**.

The product is a control plane — not one assay

Buyers purchase:

1. **Execution platform** — DomainProgram language, typed actions, cfg/wf registry, portal, gateway workers, CAAS
2. **Process packs** — omics modalities (methylation, RNA-Seq, proteomics)
3. **Application packs** — indication/trait overlays as **config** on an existing process

DISEASE-AGNOSTIC

ANALYTE-EXTENSIBLE

OPERATOR-CONFIGURABLE

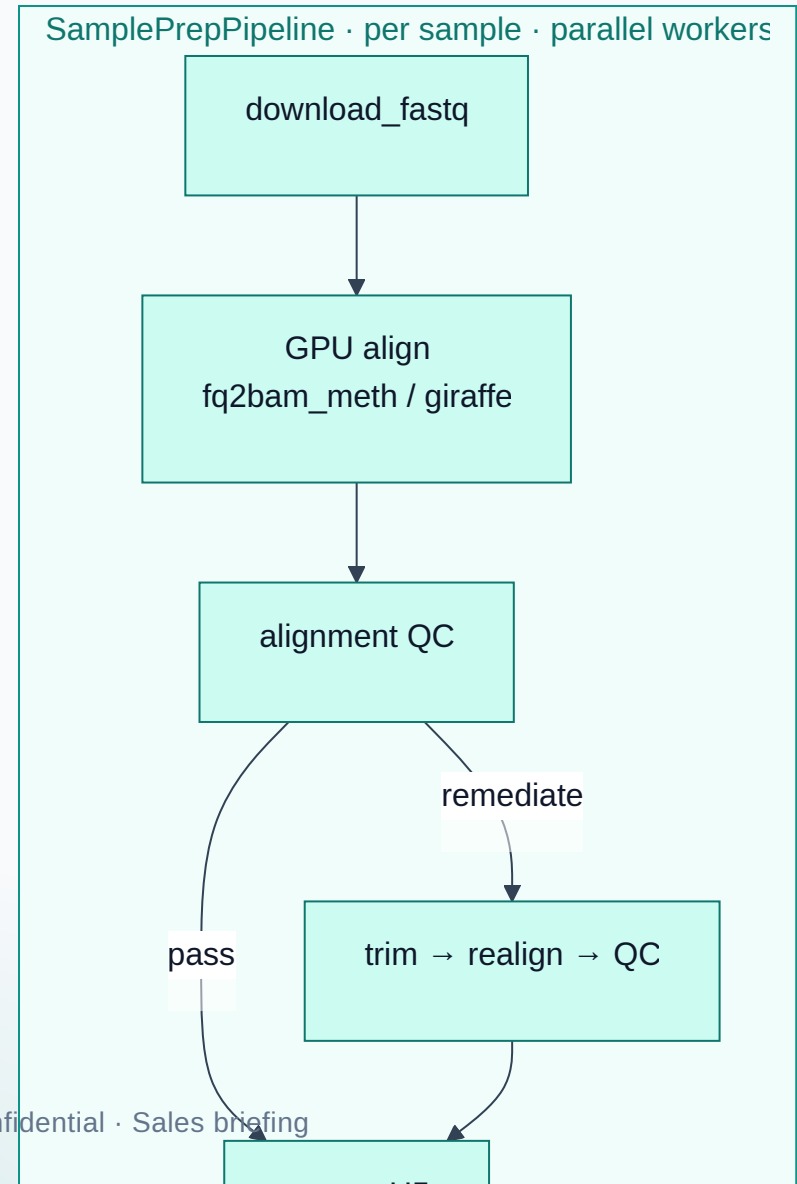
DomainProgram: declare the science once

LANGUAGE POWER

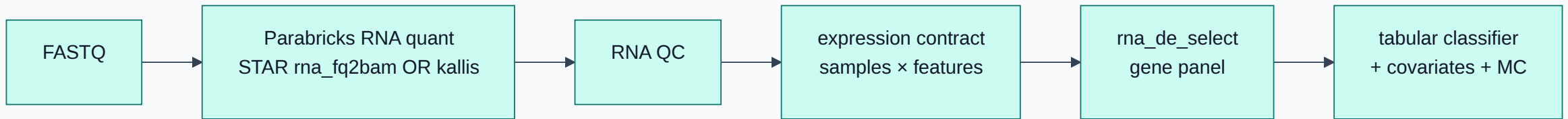
- A **DomainProgram** is a versioned, typed workflow IR — not a shell script farm.
- Compiles into schedulable graphs for **local** or **distributed** workers.
- Same language expresses SamplePrep, Monte Carlo stability, freeze, model selection, and holdout gates.
- Operators change **topology and parameters through config layers** — workers execute baked `resolvedConfig`.

Flexibility without coding chaos: new modalities add programs + typed actions on the **same** scheduler, QC gates, and database structures.

DomainProgram topology (methylation process)

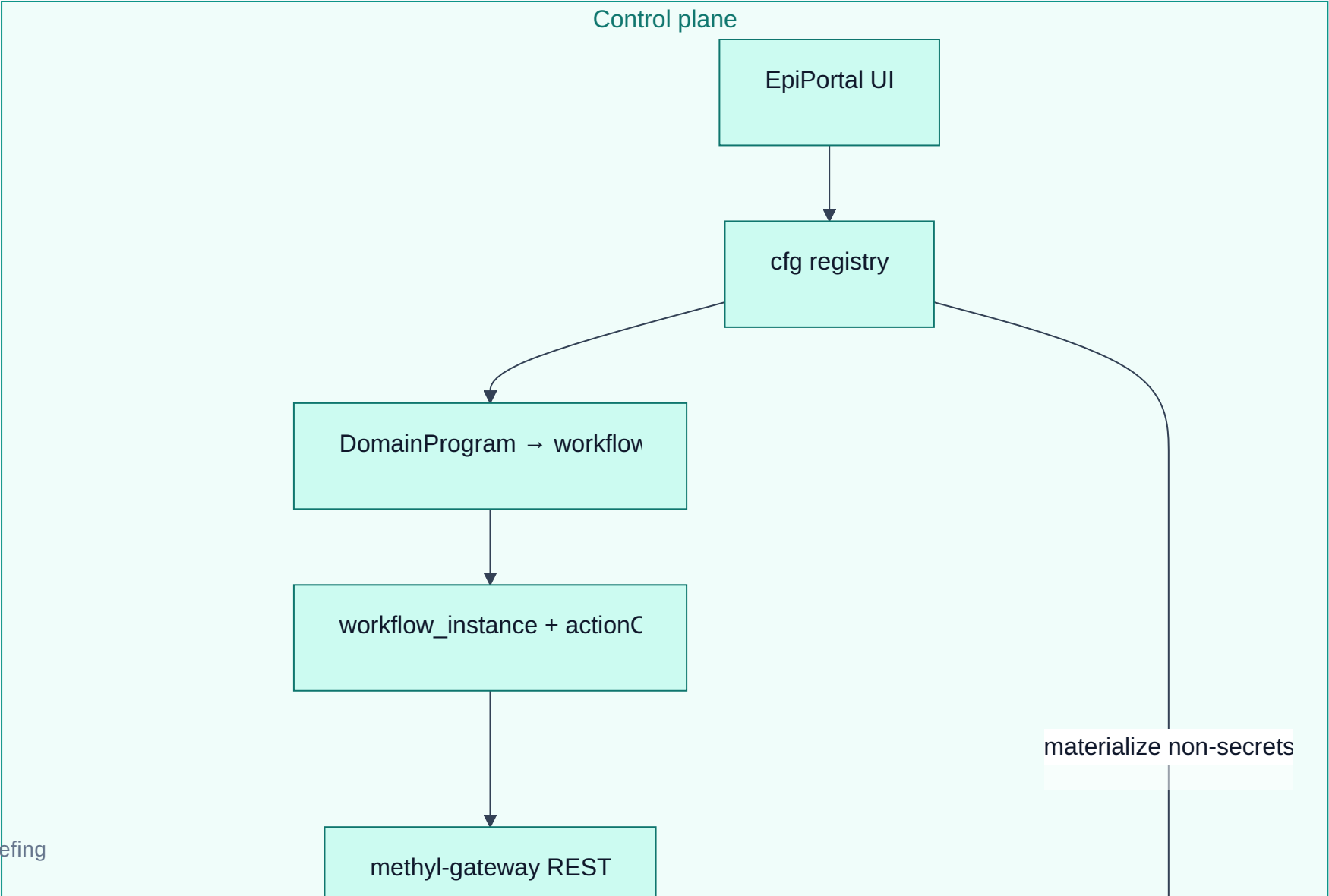


Same language, different modality (RNA-Seq process pack)

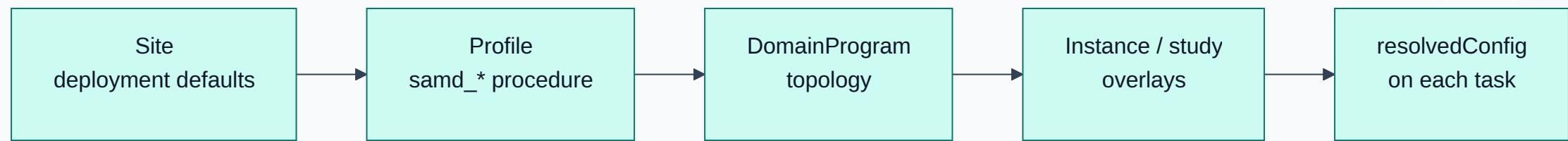


- Selects quant mode via `actionConfig.rna_align.quant_mode`.
- Reuses scheduler, cfg/wf, CAAS, and the shared feature seam — **not** a separate product stack.
- Remaining gate: representative cohort data + validation evidence (not aligner R&D).

Distributed execution: portal → gateway → workers

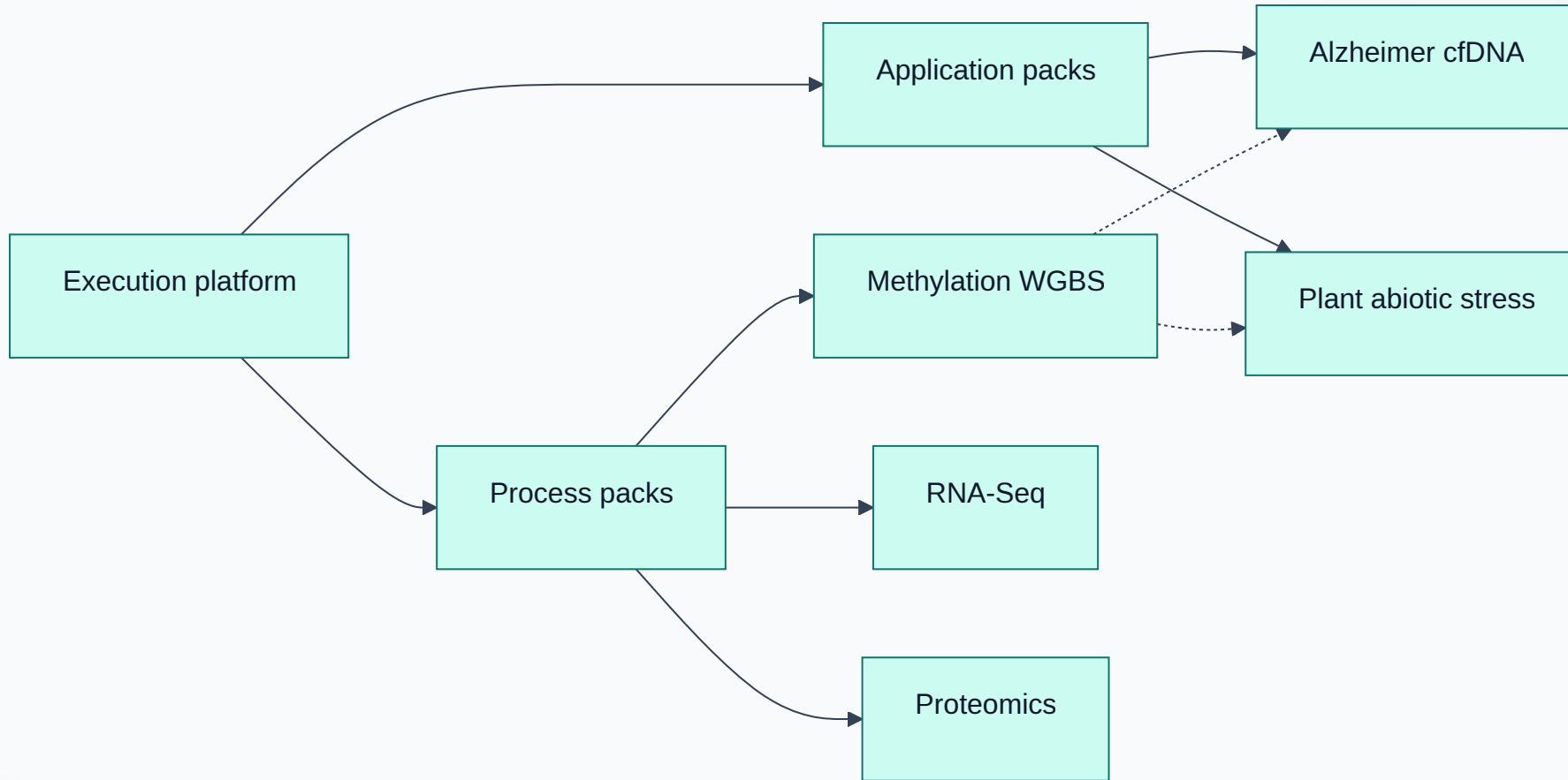


Config layers (highest wins) — tune without redeploying science code



Layer	Who edits	Examples
Site	Operators / cluster	genomes, caches, site action caps
Profile	Release / procedure pack	samd_research, MC knobs
DomainProgram	Platform / pack authors	node graph, typed actions
Study / instance	Study team	cohorts, partitions, claim gates

Process packs vs application packs



Process pack = new modality

DomainPrograms, typed actions, QC, profiles on the shared control plane.

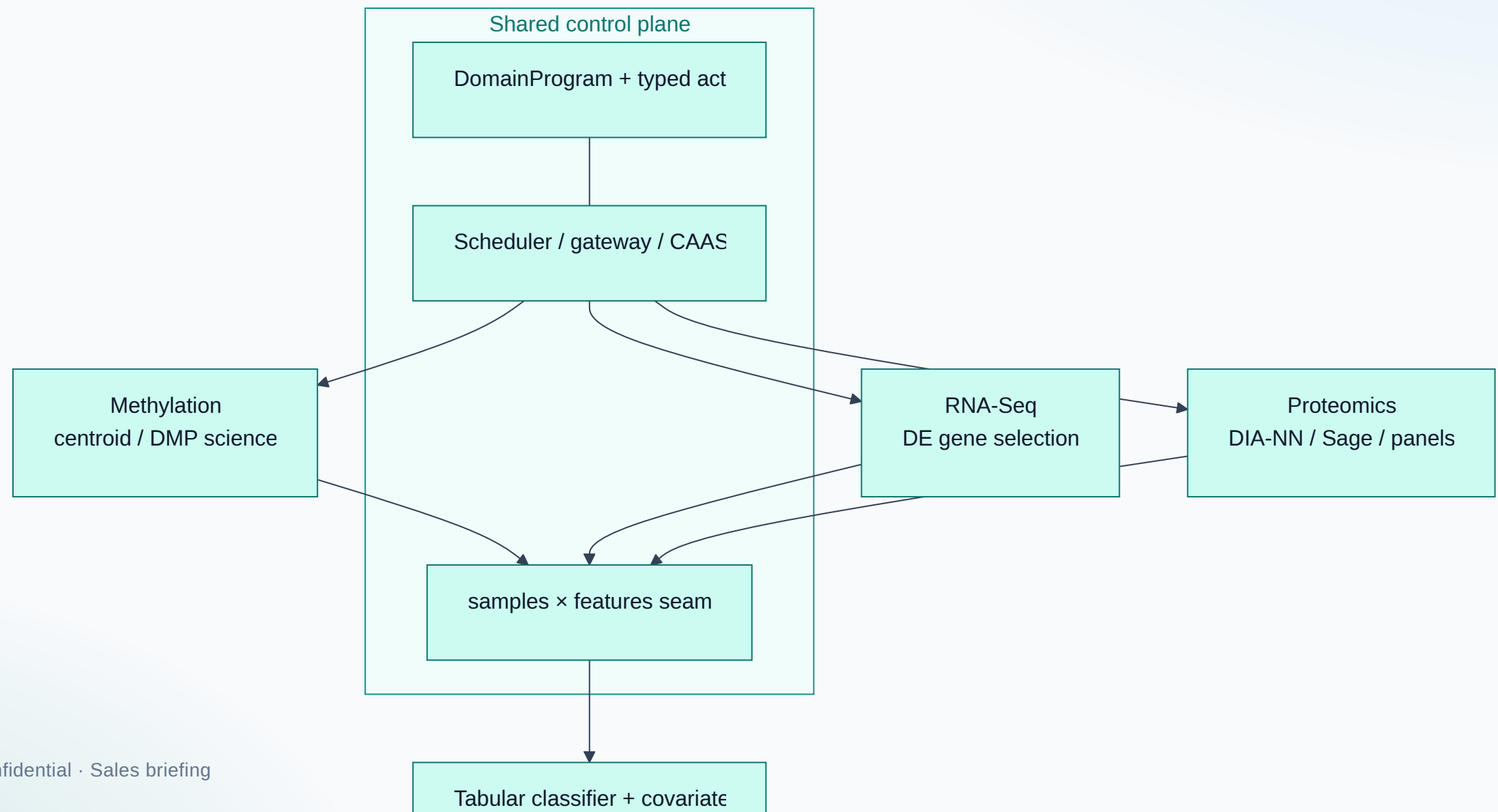
Application pack = config overlay

Study manifest, cohorts, partitions, disease/trait overlay — **no new aligner**.

Analyte & disease-process roadmap

Process / analyte	Status	What ships
Methylation — buffy / cfDNA (oncology)	In production	SamplePrep → MC stability → freeze → model; SaMD ladder
RNA-Seq	Research process pack	STAR/kallisto, RNA QC, DE panel → tabular classifier
Proteomics	Research process pack	GPU DIA-NN, Sage DDA, panel ingest; Prosit / Casanovo
Alzheimer cfDNA	Research application pack	Methylation + disease overlay (neuro-core); config only
Plant abiotic stress	Research application pack	Control vs Drought WGBS; multi-crop sites; plant-stress-core

Multiomics on one plane (why this is flexible)



Application pack pattern (config, not code)

Alzheimer cfDNA

`primary_analyte: cfdna`

Patient-disjoint partitions; `mapper.disease_term` = Alzheimer's; `neuro-core` enrichment; Control → MCI → AD progression.

No new actions or aligners.

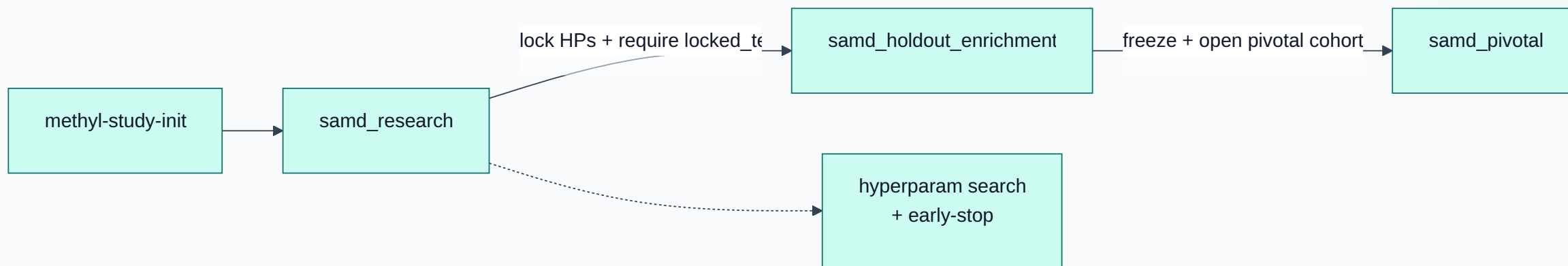
Plant abiotic stress

`primary_analyte: plant_tissue`

Binary Control vs Drought across CG/CHG/CHH; Ensembl Plants sites; optional deconv lifecycle; offline `plant_traits` prior.

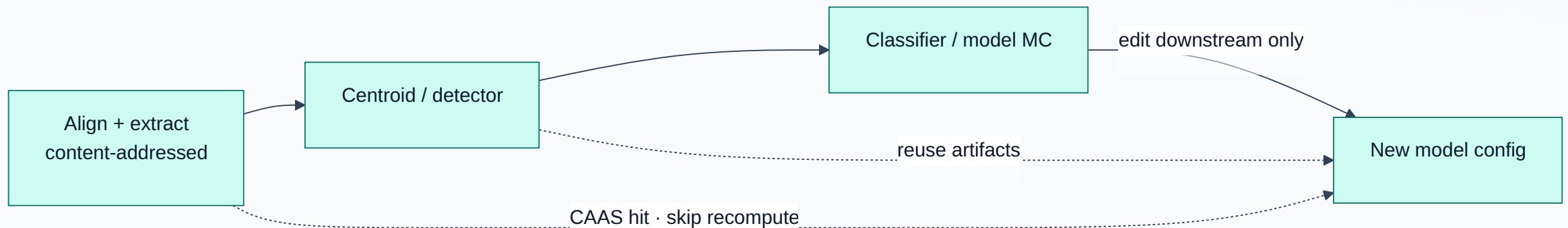
epi-GBS isolated via separate SamplePrep program.

SaMD profile ladder (research → pivotal)



- **Research:** discover stable panels under partition discipline.
- **Holdout enrichment:** locked hyperparameters + locked test patients.
- **Pivotal:** claim-gated clinical-performance evidence packaging.

CAAS: change the classifier, keep the alignment

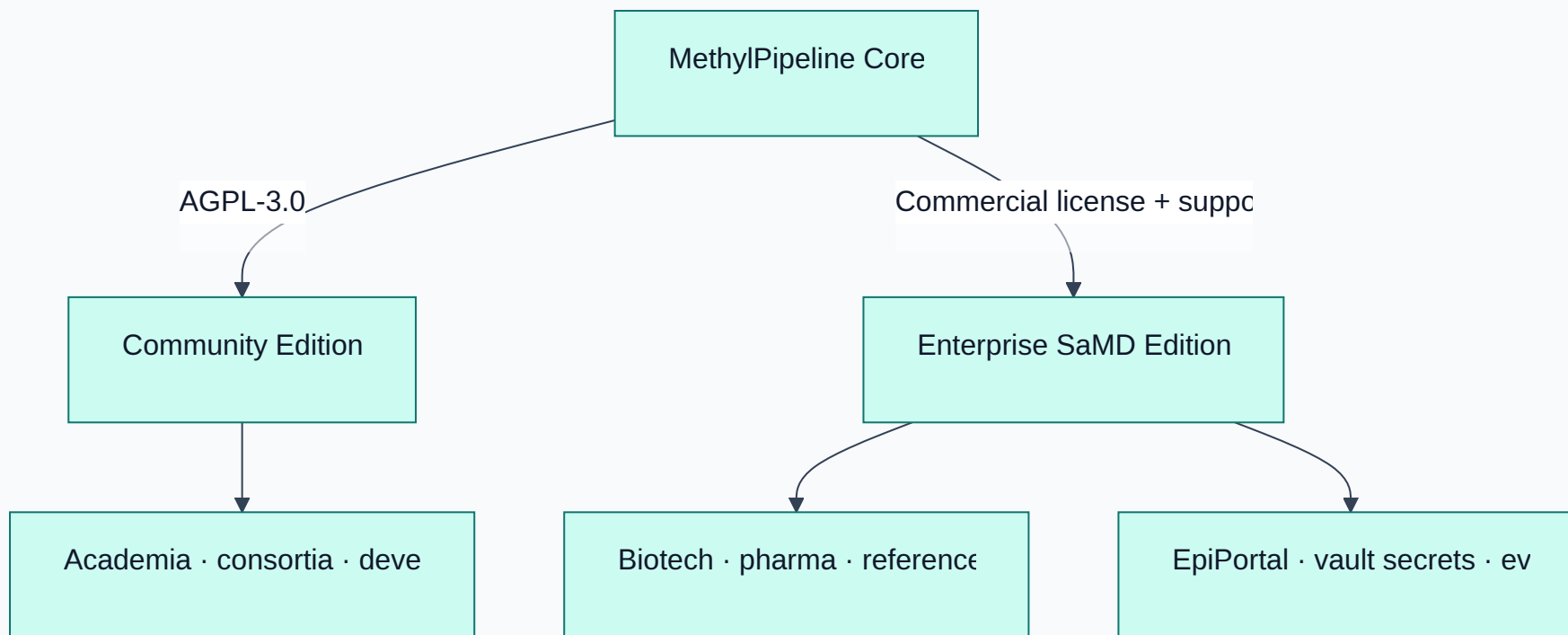


- `.action_results` bind input/output hashes, CLI versions, environment signatures.
- Large-cohort VM cost drops when science iteration stays **downstream**.
- Provenance chain is ready for quality review — not a slideware promise.

Packaging & GTM

Open core that captures R&D — Enterprise that closes the FDA chasm

Open-core packaging



Community vs Enterprise

Community Edition

AGPL-3.0

- LocalWorkflowEngine
- DNA methylation process pack
- CLI tools + `samd_research`
- Goal: citations, adoption, contributions

Enterprise / SaMD Edition

ANNUAL SUBSCRIPTION

- Evidence index + traceability packaging
- Pivotal ladder unlocks
- EpiPortal multi-tenant ops
- Zero-trust credentials + GPU SamplePrep support

Hybrid cloud (BYOC) for sensitive genomics

We host / operate

- EpiPortal control plane
- DB metadata + scheduler
- Non-secret materialization

Customer keeps

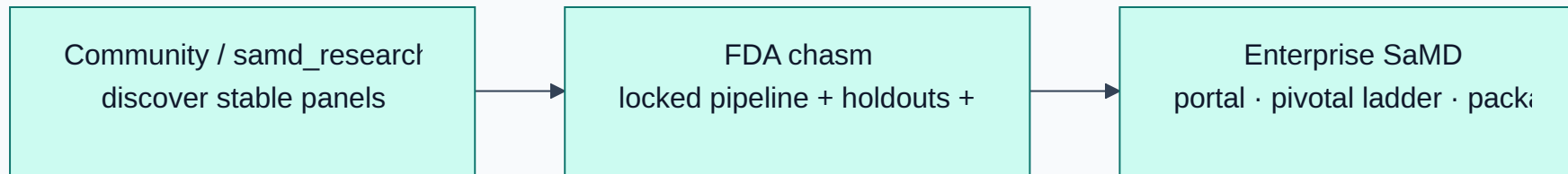
- Stateless workers on AWS / Azure / on-prem
- `/work` + samples in their boundary
- HIPAA / GDPR data residency

Bring Your Own Cloud: regulated data never has to leave the customer's estate.

Who buys — and why

1. **Biotech / diagnostic startups (primary)** — liquid biopsy & multiomics programs that need QA + regulatory framework without building a platform.
2. **Pharma / trial sponsors** — multi-year locked, reproducible workflows for methylation (and other packs) as endpoints.
3. **Reference labs / CROs** — upsell FASTQs into curated classification + evidence packages (oncology, Alzheimer cfDNA, ...).

Customer journey: the “FDA chasm”



- **Hook:** free/low-cost research feasibility.
- **Trigger:** promising marker must survive internal + pivotal validation under audit.
- **Upgrade:** pivotal ladder support, portal operations, regulatory packaging.

Why technical buyers say yes

- **Audit trails with low overhead** — hashes, versions, environment signatures in action results.
- **CAAS economics** — skip alignment/extraction across large cohorts when iterating models.
- **Multionics extensibility** — process packs add DomainPrograms; application packs are config-only overlays.
- **Partition discipline** — development vs locked holdout patients enforced by profiles, not heroics.

Close

Regulatory-Ready Platform for Multiomics Diagnostics

Control plane + packs — not a single hard-coded assay.

Next step: live DomainProgram demo · SaMD ladder walkthrough · evidence-scaffold review

